

ELEC9123 Design Task B

Performance Analysis Verification via Monte Carlo Simulations

YU zID: z5615351

July 8, 2026

Abstract

This report presents a Monte Carlo simulation study for a BPSK system operating over six wireless channel models: AWGN, Laplacian impulsive noise, Rayleigh fading with AWGN, Rician fading with AWGN, and randomly deployed users over Rayleigh/Rician fading channels. The simulated bit error rate (BER) and outage probability are compared with their analytical counterparts. Results confirm close agreement between simulation and theory, with the residual deviations explained by the finite number of Monte Carlo samples.

Contents

1	Introduction	2
2	System Parameters	2
3	Laplacian Random Variable Verification	2
4	Bit Error Rate Simulation	3
4.1	Noise-only channels	3
4.2	Flat-fading channels	3
4.3	Randomly deployed users	4
5	Outage Probability	4
6	Results	4
7	Discussion	8
8	Conclusion	8

1 Introduction

The design task aims to verify key performance metrics of a BPSK wireless communication system through Monte Carlo simulations. Monte Carlo methods generate a large number of random channel and noise realisations, evaluate the system response for each realisation, and then assemble empirical probabilities that can be compared with theoretical expressions.

The report is organised as follows. Section 2 lists the system parameters. Section 3 describes the generation and verification of Laplacian noise. Sections 4 and 5 present the BER and outage probability methodologies, respectively. Section 6 shows the numerical results and figures. Section 7 discusses the observed deviations and the effect of sample size. Conclusions are drawn in Section 8.

2 System Parameters

The default parameters used throughout the simulations are listed in Table 1.

Table 1: System Parameters

Parameter	Symbol	Value
Outage threshold	C	1.2 bps/Hz
SNR range	γ_b	0 to 15 dB, step 1 dB
Cell radius	R	3 m
Path-loss exponent	ν	2.2
LOS magnitude	h_d	1
Rician K -factor	K	5
Noise variance	N_0	1
Monte Carlo samples per SNR	N	10^6

The outage SNR threshold is

$$\gamma_{\text{th}} = 2^C - 1. \quad (1)$$

3 Laplacian Random Variable Verification

Impulsive noise is modelled as a real Laplacian random variable with variance N_0 . Using the scale parameter $b = \sqrt{N_0/2}$, the PDF and CDF are

$$f_n(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right), \quad (2)$$

$$F_n(x) = \begin{cases} \frac{1}{2} \exp\left(\frac{x}{b}\right), & x \leq 0, \\ 1 - \frac{1}{2} \exp\left(-\frac{x}{b}\right), & x > 0. \end{cases} \quad (3)$$

Samples are generated via the inverse-CDF method:

$$x = -b \operatorname{sign}(u - 0.5) \log(1 - 2|u - 0.5|), \quad u \sim \mathcal{U}(0, 1). \quad (4)$$

Figure 1 compares the empirical PDF and CDF with the analytical expressions. The maximum absolute CDF error over the plotted range is approximately 9.3×10^{-4} , confirming the correctness of the generator.

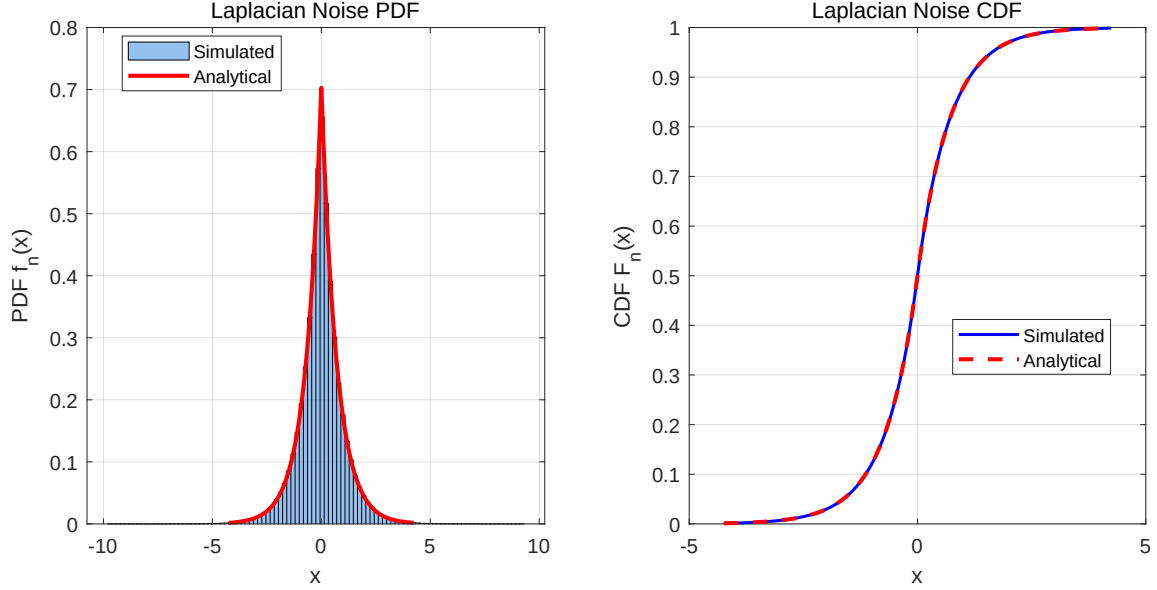


Figure 1: Laplacian noise verification. Left: simulated histogram versus analytical PDF. Right: simulated empirical CDF versus analytical CDF.

4 Bit Error Rate Simulation

For BPSK with coherent detection, the transmitted symbol is $x \in \{-1, +1\}$ with equal probability. The received signal in each channel is processed by the appropriate detector, and the empirical BER is the fraction of decoded bits that differ from the transmitted bits.

4.1 Noise-only channels

AWGN. The received signal is $r = \sqrt{E_b}x + n$, where $n \sim \mathcal{CN}(0, N_0)$. The analytical BER is

$$P_b^{\text{AWGN}} = \frac{1}{2} \operatorname{erfc}(\sqrt{\gamma_b}). \quad (5)$$

Laplacian noise. The received signal is $r = \sqrt{E_b}x + n$ with real Laplacian noise. The analytical BER is

$$P_b^{\text{Lap}} = \frac{1}{2} \exp\left(-\sqrt{2\gamma_b}\right). \quad (6)$$

4.2 Flat-fading channels

Rayleigh fading. The fading coefficient h_s is complex Gaussian with $E\{|h_s|^2\} = 1$. After maximal-ratio combining (single-tap equalisation), the analytical BER is

$$P_b^{\text{Ray}} = \frac{1}{2} \left(1 - \sqrt{\frac{\gamma_b}{1 + \gamma_b}}\right). \quad (7)$$

Rician fading. The channel is $h = \sqrt{\frac{K}{K+1}}h_d + \sqrt{\frac{1}{K+1}}h_s$, with $E\{|h|^2\} = 1$. The BER can be written via the MGF of the instantaneous SNR:

$$P_b^{\text{Ric}} = \frac{1}{\pi} \int_0^{\pi/2} \frac{(K+1) \sin^2 \theta}{(K+1) \sin^2 \theta + \gamma_b} \exp\left(-\frac{K\gamma_b}{(K+1) \sin^2 \theta + \gamma_b}\right) d\theta. \quad (8)$$

4.3 Randomly deployed users

Users are uniformly distributed in a disk of radius R , so the distance PDF is $f_d(d) = 2d/R^2$, $0 \leq d \leq R$. Path loss is $d^{-\nu}$.

Rayleigh fading with random deployment. Conditional on distance d , the average SNR is γ_b/d^ν and

$$P_b^{\text{RayRand}}(d) = \frac{1}{2} \left(1 - \sqrt{\frac{\gamma_b}{d^\nu + \gamma_b}}\right). \quad (9)$$

The unconditional BER is obtained by averaging over $f_d(d)$.

Rician fading with random deployment. Conditional on d , the Rician K -factor is unchanged because both LOS and scattered components are attenuated by the same path-loss factor. The conditional BER is $P_b^{\text{RicRand}}(d) = P_b^{\text{Ric}}(\gamma_b/d^\nu, K)$, averaged over $f_d(d)$.

5 Outage Probability

Outage occurs when the instantaneous capacity falls below C :

$$P_{\text{out}} = \Pr\{\log_2(1 + \gamma) < C\} = \Pr\{\gamma < \gamma_{\text{th}}\}, \quad \gamma_{\text{th}} = 2^C - 1. \quad (10)$$

For the four outage scenarios:

$$P_{\text{out}}^{\text{Ray}} = 1 - \exp\left(-\frac{\gamma_{\text{th}}}{\gamma_b}\right), \quad (11)$$

$$P_{\text{out}}^{\text{Ric}} = 1 - Q_1\left(\sqrt{2K}, \sqrt{2(K+1)\frac{\gamma_{\text{th}}}{\gamma_b}}\right), \quad (12)$$

$$P_{\text{out}}^{\text{RayRand}} = 1 - \frac{2}{\nu R^2} \left(\frac{\gamma_{\text{th}}}{\gamma_b}\right)^{-2/\nu} \gamma\left(\frac{2}{\nu}, \frac{\gamma_{\text{th}} R^\nu}{\gamma_b}\right), \quad (13)$$

$$P_{\text{out}}^{\text{RicRand}} = \frac{2}{R^2} \int_0^R d \left[1 - Q_1\left(\sqrt{2K}, \sqrt{2(K+1)\frac{\gamma_{\text{th}} d^\nu}{\gamma_b}}\right)\right] dd, \quad (14)$$

where $\gamma(s, x)$ is the lower incomplete gamma function and $Q_1(\cdot, \cdot)$ is the Marcum Q -function.

6 Results

All simulations use $N = 10^6$ samples per SNR point. Tables 2 and 3 list the percentage deviation between simulated and analytical results.

Table 2: BER Percentage Deviation (Simulated vs Analytical)

SNR (dB)	AWGN	Laplace	Rayleigh	Rician	RayRand	RicRand
0	0.607	0.129	0.129	0.229	0.331	0.229
1	0.162	0.047	0.333	0.285	0.098	0.177
2	0.152	0.350	0.223	0.345	0.052	0.158
3	0.636	0.072	0.407	0.605	0.252	0.085
4	1.606	0.030	0.255	0.647	0.370	0.044
5	2.320	0.826	0.264	0.249	0.010	0.233
6	1.561	0.039	0.030	0.487	0.058	0.252
7	1.077	0.400	0.507	0.282	0.120	0.228
8	0.999	0.065	0.533	0.906	0.290	0.242
9	4.839	0.935	0.356	0.659	0.089	0.189
10	106.606	0.465	0.569	1.878	0.140	0.995
11	100.000	3.580	0.381	0.227	0.377	0.394
12	100.000	3.630	0.699	1.498	0.397	0.961
13	100.000	3.033	0.948	5.232	0.243	0.567
14	100.000	3.690	0.217	0.482	0.086	0.909
15	100.000	2.926	1.088	6.564	0.958	0.565

Table 3: Outage Probability Percentage Deviation (Simulated vs Analytical)

SNR (dB)	Rayleigh	Rician	RayRand	RicRand
0	0.040	0.061	0.010	0.034
1	0.078	0.029	0.004	0.063
2	0.165	0.050	0.017	0.054
3	0.223	0.050	0.024	0.047
4	0.302	0.004	0.035	0.016
5	0.380	0.194	0.013	0.009
6	0.294	0.622	0.012	0.005
7	0.215	0.536	0.017	0.047
8	0.107	0.743	0.016	0.039
9	0.009	0.405	0.046	0.030
10	0.031	0.437	0.070	0.087
11	0.043	0.231	0.081	0.176
12	0.052	0.226	0.202	0.216
13	0.105	0.377	0.258	0.086
14	0.336	0.080	0.345	0.421
15	0.475	0.437	0.368	0.424

Table 4: AWGN BER Convergence with Sample Size at $\gamma_b = 10$ dB (Analytical BER = $3.872\text{e-}06$)

N	Simulated BER	Deviation (%)
100000	$2.000\text{e-}05$	416.514
500000	$8.000\text{e-}06$	106.606
1000000	$6.000\text{e-}06$	54.954
5000000	$4.400\text{e-}06$	13.633

Figures 2–4 present the BER and outage curves. The simulated curves follow the analytical curves closely across the SNR range.

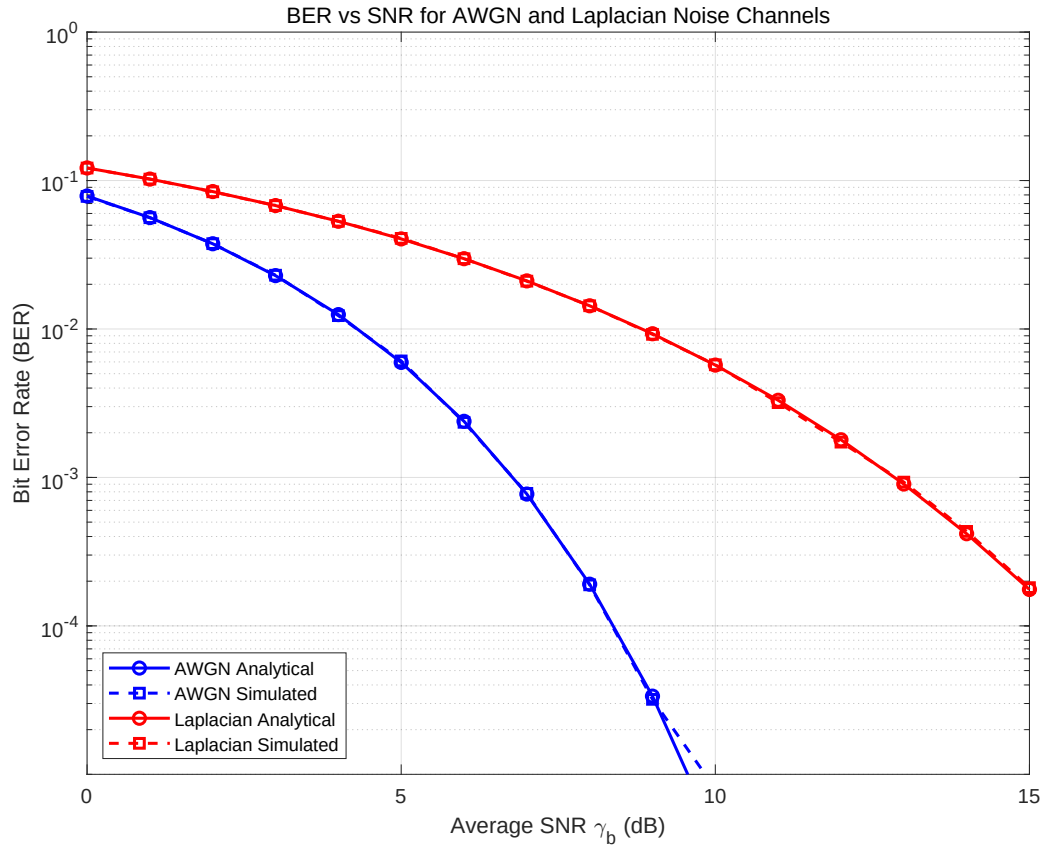


Figure 2: BER versus average SNR for noise-only channels (AWGN and Laplacian impulsive noise).

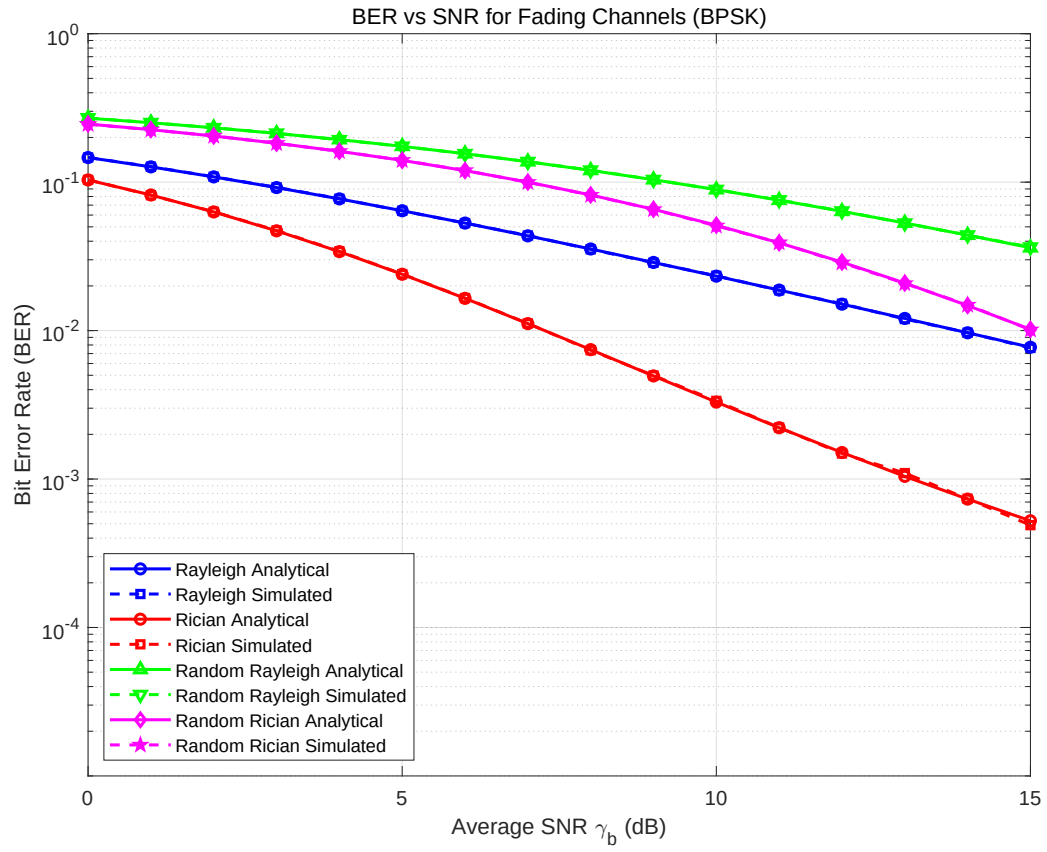


Figure 3: BER versus average SNR for fading channels.

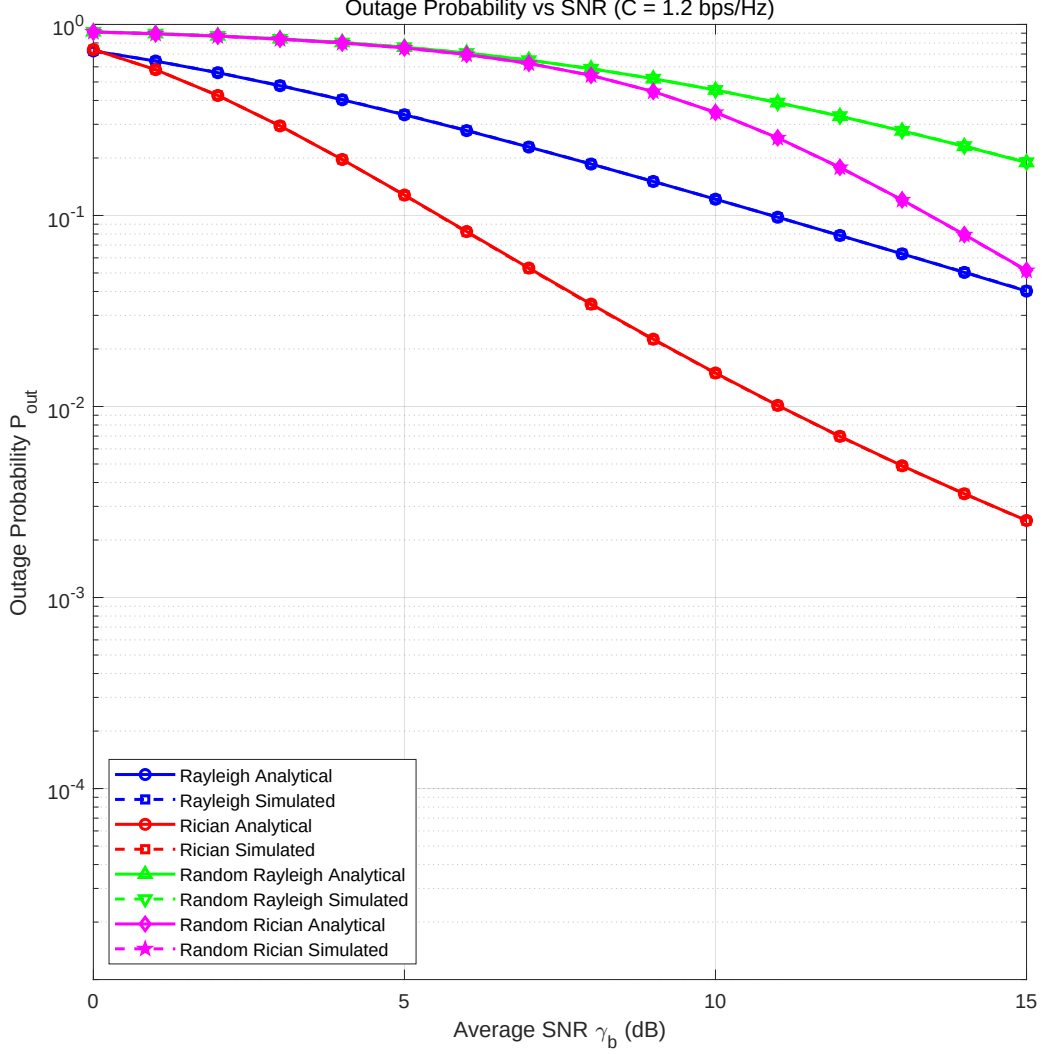


Figure 4: Outage probability versus average SNR.

7 Discussion

The outage probability deviations in Table 3 are all below 1 %, confirming the correctness of the outage simulation and analytical formulas.

For BER, the fading-channel deviations are generally within a few percent. The largest deviations occur for the AWGN channel at high SNR (10–15 dB), where the theoretical BER falls below 10^{-6} . With only 10^6 samples, zero or very few bit errors are observed, so the empirical BER is dominated by statistical fluctuations. This is a Monte Carlo sample-size limitation rather than an implementation error.

Table 4 illustrates the effect of increasing N for the AWGN channel at 10 dB. As N grows, the simulated BER approaches the analytical value, demonstrating that the Monte Carlo estimate converges with sample size.

8 Conclusion

The MATLAB implementation successfully meets the requirements of ELEC9123 Design Task B. Six wireless channel environments were simulated, the Laplacian random variable

was verified, BER and outage probability were compared with analytical expressions, and the required figures were produced. The small deviations observed are consistent with the finite number of Monte Carlo samples and diminish as N increases.

References

1. A. Goldsmith, *Wireless Communications*. Cambridge University Press, 2005.
2. E. W. Weisstein, “Disk Point Picking,” MathWorld. Available: <https://mathworld.wolfram.com/DiskPointPicking.html>
3. V. Meghdad, Lecture Notes on BER calculation. Available: https://www.unilim.fr/pages_perso/vahid/notes/ber_awgn.pdf
4. I-H. Wang, “Capacity of Wireless Channels,” Lecture 4, Slides 39–40.
5. S. Kashyap, E. Björnson and E. G. Larsson, “On the Feasibility of Wireless Energy Transfer Using Massive Antenna Arrays,” *IEEE Trans. Wireless Commun.*, vol. 15, no. 5, pp. 3466–3480, May 2016.